

Affective visualization graphs

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Setting up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization -- dont need...
library(janitor) # cleaning data frames
library(DHARMA) # for diagnostics
```

```
# read in personal data to object from csv
my_data <- read_csv("envs193ds-personal-data.csv")
```

Cleaning and wrangling data

```

# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
    workout_type == "Run" ~ 1, # run = 1
    .default = 0               # all other workout types = 0
  ))

```

Fitting model

```

# fitted generalized linear model
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_glm <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)

```

Diagnostics

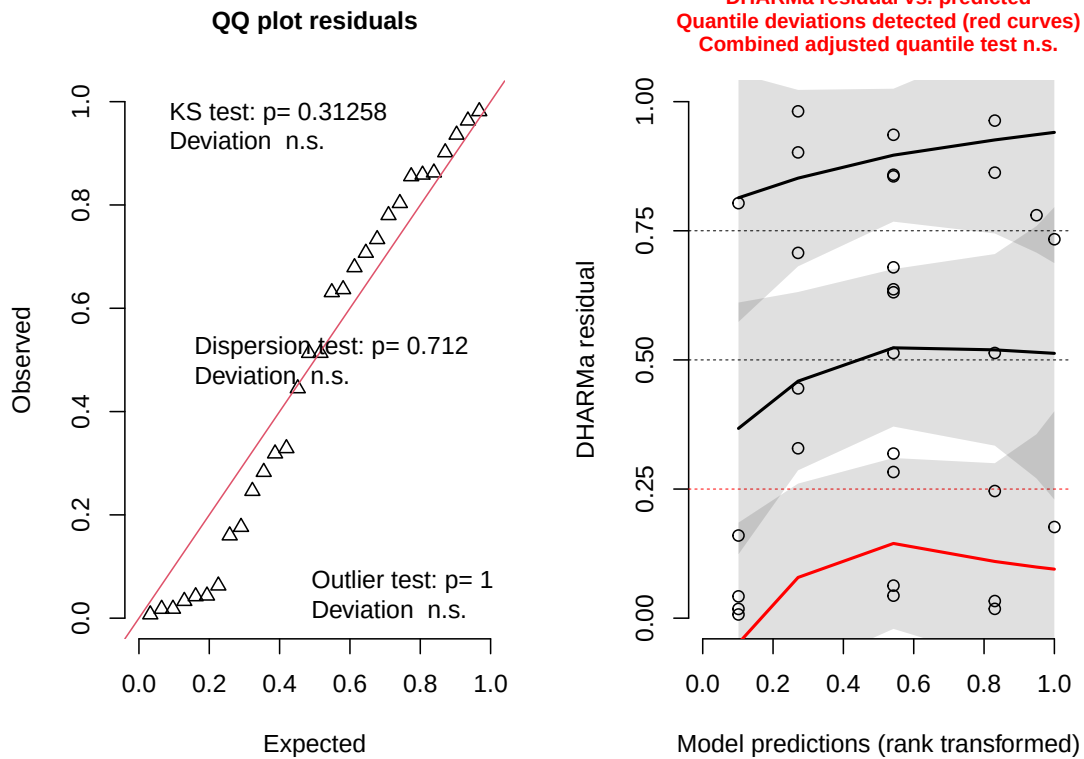
```

# simulated residuals from GLM for diagnostic plots
run_simres <- simulateResiduals(run_glm)

# plotted DHARMA diagnostics
# left plot: KS test - do residuals follow uniform distribution?
# right plot: residuals vs predicted values
plot(run_simres)

```

DHARMA residual



Coefficients

```
# displayed model summary
summary(run_glm)
```

Call:

```
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.7011	3.4964	-2.489	0.0128 *
sleep_hours	1.0873	0.4712	2.307	0.0210 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.652 on 29 degrees of freedom
Residual deviance: 28.046 on 28 degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5

```
# exponentiated coefficients to get odds ratios  
# e.g. exp(coef) = odds ratio for 1 hr increase in sleep  
exp(coef(run_glm))
```

```
(Intercept)  sleep_hours  
0.0001663954  2.9663093920
```

```
# exponentiated confidence intervals  
exp(confint(run_glm))
```

```
                2.5 %      97.5 %  
(Intercept) 3.108829e-08 0.04889274  
sleep_hours  1.372042e+00 9.39665076
```